Assignment No: 20

Problem Statement: Write a C function to swap two numbers with and without using pointers.

Objective:
- To understand the concept of function calls in the C programming language.
- To learn the difference betn call by value and call by reference.

Theory: In the C programming language, swapping two numbers can be performed in two ways: without pointers and with pointers. When swapping numbers without pointers, the values are passed to the function by value.

Platform: Linux.

Input: Two integer numbers.

Output: Displays swapped values using both call by value and call by reference methods.

Sample Input: Enter two numbers: 10 20

Sample Output: After swapping without pointers.
a=20 b=10
After swapping with pointers:
x=20 y=10

Conclusion: Thus, the C program successfully demonstrates the difference between swapping values using call by value and call by reference

FAQs:

1) What is difference between call by value & call by reference in C?

A. Call by Value: a) A copy of actual value is passed to the function.
b) Changes made inside the function do not affect the original variable.
Call by Reference: a) The address of the variable is passed
b) Changes made inside the function do affect the original variable.

2) How do pointers help modify original variables inside a function?

A. a) Pointers allow a function to directly access and modify the original variables' memory location, instead of working on a copy.
b) Every variable is stored in memory with an address.

3) Why is swapping using pointers more effective in certain programs?

A. Swapping using pointers is often more effective because it works directly on the original variables instead of copies.